

Brain Tumor MRI Classification

An Educational Prototype for Multi-Class Neuroimaging Classification

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1 Introduction and Problem Statement

1.1 Clinical Motivation

Brain tumors are among the most clinically consequential findings in neuroimaging. Accurate classification can help guide treatment planning and give patients more time to explore treatment opportunities in early stages. Manual MRI classification relies on radiologist review, which is both time-intensive and subject to some level of human error. Computer-aided detection can help support this review process, with deep learning approaches demonstrating strong performance on benchmark datasets in recent literature [1].

1.2 Four-Class Problem

The dataset targets four mutually exclusive categories of brain MRI scan: glioma, meningioma, pituitary, and no tumor.

- **Glioma** refers to tumors arising from glial cells; they are the most common primary brain malignancy and can appear throughout the cerebral hemispheres, making their MRI presentation highly variable [2].
- **Meningioma** originates in the meninges — the protective membranes surrounding the brain and spinal cord — and typically presents as a well-defined mass near the cortical surface; it is usually benign but can cause significant neurological symptoms due to compression [2].
- **Pituitary tumors** arise in the pituitary gland at the base of the skull and are generally small and anatomically localized, making them relatively distinguishable in imaging [2].
- **No tumor** refers to scans with no detectable pathology and serves as the negative class.

1.3 Project Scope and Framing

This project is an educational prototype. Its purpose is to demonstrate the application of a progression of ML methods to a real medical imaging dataset, and to report results with appropriate honesty about limitations and failure modes. The models produced here are not intended for clinical use. Deployment of any automated diagnostic tool would require prospective clinical validation, patient-level data provenance, demographic diversity testing, and regulatory approval, all of which are outside the scope of this project. Additionally, the dataset used provides no patient identifiers, so patient-level independence between train and test sets cannot be verified meaning reported metrics may therefore be optimistic.

2 Data Description

2.1 Dataset Source

The **Brain Tumor MRI Dataset (V2)** by Masoud NickParvar [3] is a curated benchmark merging three public sources: Figshare, SARTAJ, and Br35H. Labels are encoded by folder name (**glioma**, **meningioma**, **notumor**, **pituitary**). The creator reports that duplicates were removed, classes were balanced, and that there is no train/test image overlap. We independently confirmed zero duplicate images across splits via MD5 hash check. The dataset provides no patient identifiers, so patient-level independence between images cannot be verified. This is acknowledged as a central limitation in Section 7.

2.2 Class Distribution

The dataset contains 7,200 images supplied as a predefined 5,600-image training folder and a 1,600-image test folder. The four classes are perfectly balanced at 1,800 images each (1,400 train + 400 test), as shown in Table 1. Perfect balance fixes the majority-class accuracy floor at exactly 25% and makes precision, recall, and F1 directly comparable across classes without reweighting.

Table 1: Class distribution in the full dataset.

Class	Train	Test	Total
Glioma	1,400	400	1,800
Meningioma	1,400	400	1,800
No Tumor	1,400	400	1,800
Pituitary	1,400	400	1,800
Total	5,600	1,600	7,200

2.3 Preprocessing Pipeline

2.3.1 Image Standardization

All images are converted to grayscale and resized to 128×128 pixels using LANCZOS resampling, then normalized to $[0, 1]$ by dividing by 255.

2.3.2 Train / Validation / Test Split

The 1,600-image test folder is held out for a single final evaluation and is not used at any point during model development. The 5,600 training images are split 80/20 with stratification (random seed 42) into 4,480 train-fit and 1,120 validation images.

2.3.3 Feature Extraction for Classical Models

For logistic regression on raw pixels, each 128×128 image is flattened to a 16,384-dimensional vector and standardized with a `StandardScaler` fit on the train-fit split only.

3 Exploratory Analysis

3.1 Class Balance

Figure 1 confirms that the dataset is perfectly balanced across all four classes in both the training and test splits, with 1,400 and 400 images per class respectively. This balance means no class-weighting is required and the 25% majority baseline is exact.

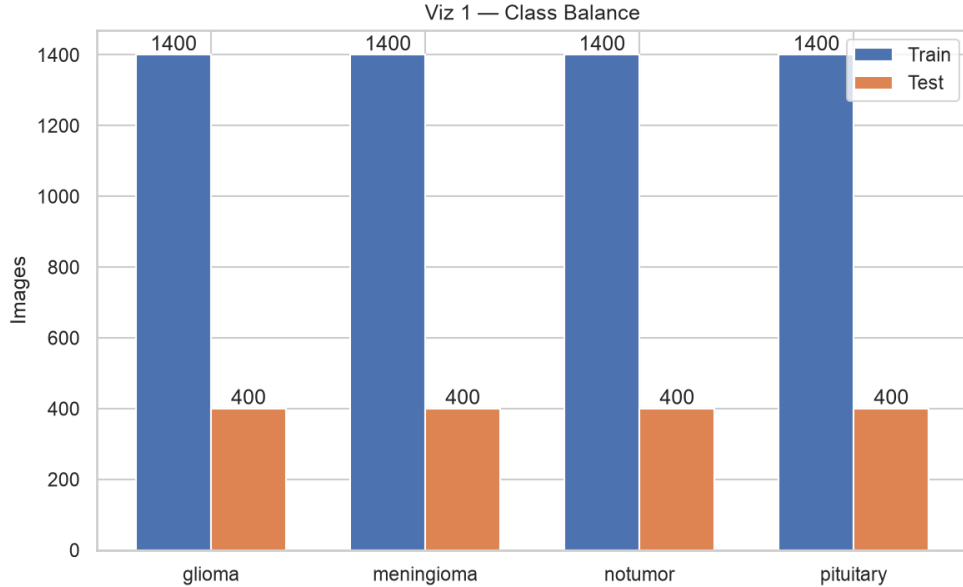


Figure 1: Class balance across train and test splits.

3.2 Dataset Grouping

Figure 2 provides a visual overview of the four-class structure of the dataset. At thumbnail scale, glioma and meningioma images are visually similar, which is a pattern consistent with the glioma–meningioma confusion observed later in all model outputs.

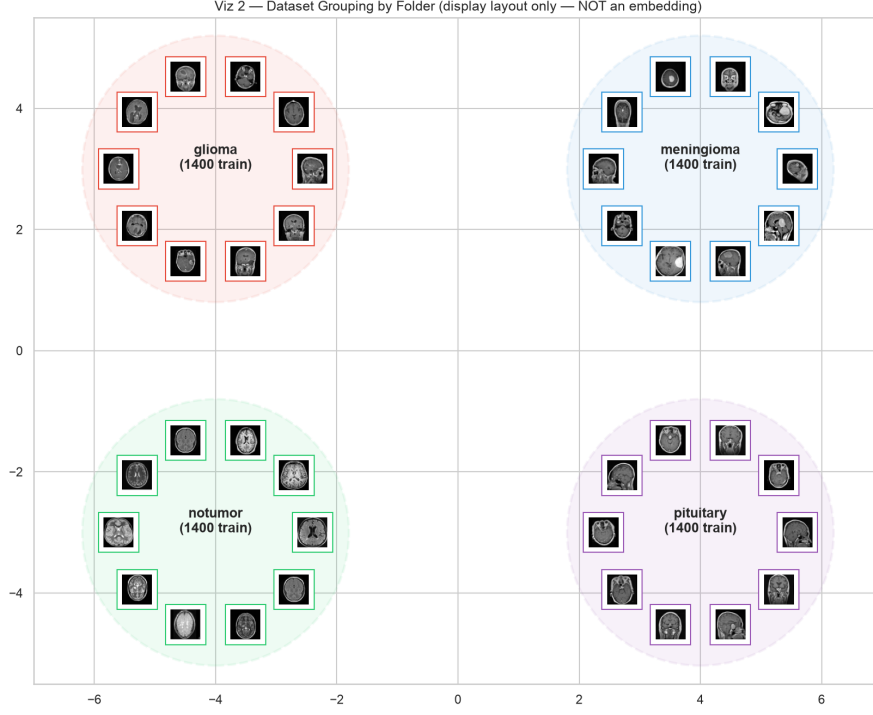


Figure 2: Representative grouping of MRI images by class.

3.3 Image Dimensions

Raw image dimensions vary considerably across the dataset, with widths ranging from 150 to 1,375 pixels and heights from 167 to 1,446 pixels across 447 unique size combinations (Figure 3). No-tumor scans are on average the smallest and show the widest size spread. Resizing all images to a uniform 128×128 is therefore mandatory before modeling.

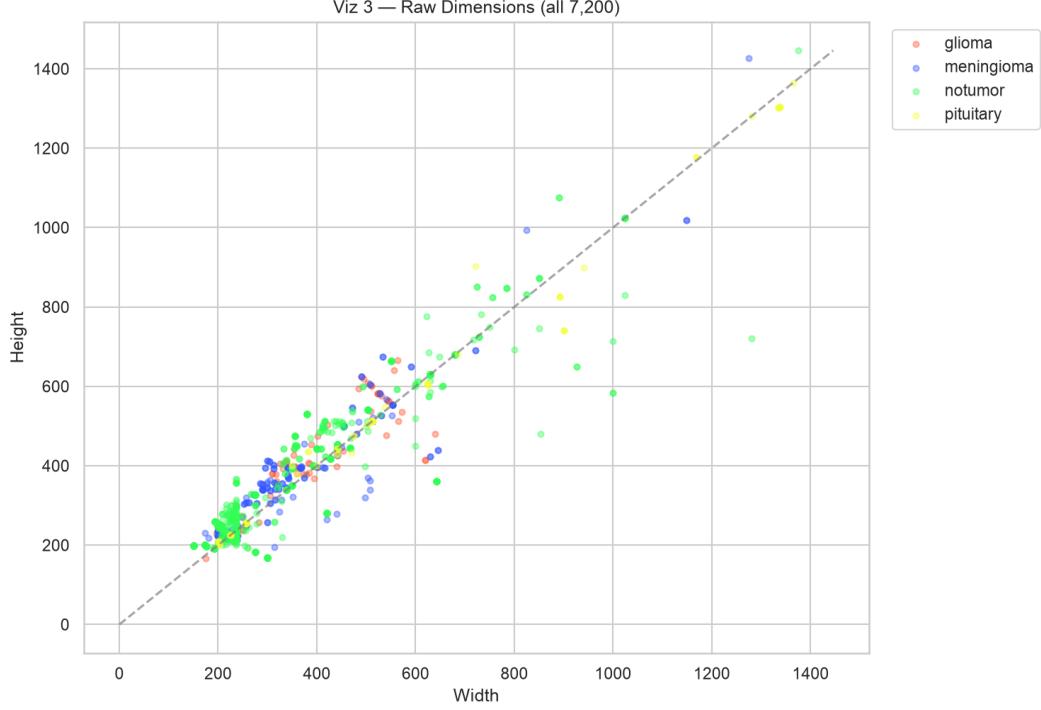


Figure 3: Raw image dimension scatter by class. No-tumor scans are systematically smaller than the tumor classes.

3.4 Brightness and Contrast

Figure 4 shows per-class pixel intensity distributions after resizing and normalization. Brightness histograms overlap substantially across tumor types, confirming that intensity alone is insufficient for four-class subtype discrimination. A single brightness threshold achieves 0.857 binary F1 for tumor vs. no-tumor detection but cannot resolve glioma, meningioma, and pituitary subtypes.

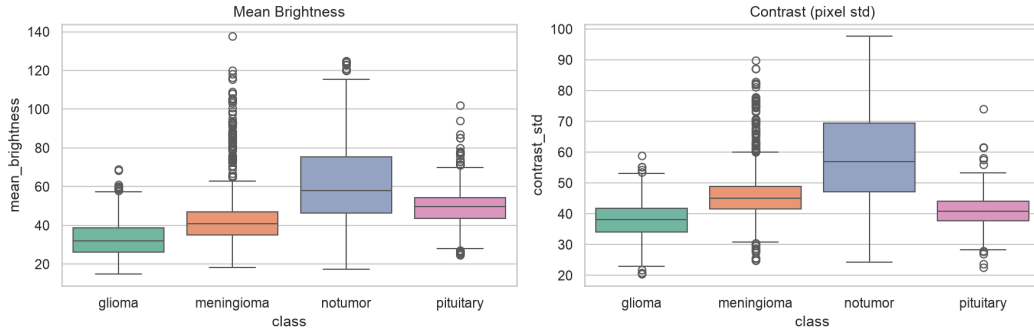


Figure 4: Pixel intensity distributions by class. Overlap across tumor types rules out brightness as a reliable classifier feature.

4 Baseline Method

4.1 Majority Class Baseline

A majority-class classifier assigns every sample to the most frequent training class regardless of input. Because the dataset is perfectly balanced, all four classes are equally frequent and the classifier defaults to a single arbitrary class, producing exactly 25% accuracy on both validation and test sets. Any learned model that does not clear this 25% threshold has extracted no useful information from the data.

On the 1,120-image validation set the majority baseline achieves 0.250 accuracy, macro precision 0.062, recall 0.250, and F1 0.100. The low macro precision reflects the fact that the classifier is never correct on three of the four classes.

5 Improved Methods

5.1 Logistic Regression on Raw Pixels

5.1.1 Architecture

Each 128×128 grayscale image is flattened to a 16,384-dimensional vector and standardized with a `StandardScaler`. A multinomial logistic regression model with softmax output is trained directly on these pixel features.

5.1.2 Training Configuration

Optimization uses the `saga` solver with $C = 1.0$ (no regularization penalty) and a maximum of 1,000 iterations. The scaler is fit on the train-fit split only.

5.1.3 Results

On the 1,120-image validation set the model achieves 0.846 accuracy, macro F1 0.844, and val log-loss 1.176, with 173 misclassified images (15.4%). On the held-out test set it achieves 0.777 accuracy and macro F1 0.776. This is a 6.9% point validation-to-test drop, indicating some overfitting to the training distribution.

5.2 Scratch CNN

5.2.1 Architecture

The scratch CNN is a sequential model with three convolutional blocks followed by a fully connected head. Each block consists of a `Conv2D` layer with ReLU activation, `BatchNormalization`, and `MaxPooling2D`. Filter counts increase from 32 to 64 to 128 across the three blocks. The head applies `GlobalAveragePooling2D`, a 64-unit dense layer with ReLU, `Dropout(0.3)`, and a 4-unit softmax output layer. Input shape is $(128 \times 128 \times 1)$.

5.2.2 Training Configuration

The model is compiled with the Adam optimizer (learning rate 10^{-3}) and sparse categorical cross-entropy loss. Training runs for up to 40 epochs with batch size 16. Early stopping monitors

validation loss with patience 8 and restores best weights. Augmentation (random horizontal flip, rotation up to 5°) is applied to the train-fit split only.

5.2.3 Results

The scratch CNN achieves 0.874 validation accuracy, macro F1 0.877, and val log-loss 0.356, with 141 misclassified images (12.6%). However, training curves reveal erratic validation loss oscillations despite smooth training loss decrease. This is a common instability of BatchNorm at small batch sizes, and is addressed in Section 5.4.

5.3 MobileNetV2 (Transfer Learning)

5.3.1 Pretrained Backbone

MobileNetV2 [4] is initialized with ImageNet weights at input shape $(128 \times 128 \times 3)$. Grayscale images are converted to three-channel tensors by stacking the single channel three times. Inputs are passed through `mobilenet_v2.preprocess_input` before the backbone.

5.3.2 Training Configuration

Training proceeds in two stages. In Stage 1 the convolutional base is frozen and only the classification head is trained: a `GlobalAveragePooling2D` layer, `Dropout(0.2)`, and a 4-unit dense output layer, compiled with Adam (10^{-4}) and `SparseCategoricalCrossentropy(from_logits=True)`. In Stage 2 the last 50 layers of the base are unfrozen and the full model is fine-tuned at a reduced learning rate of 10^{-5} . Both stages use batch size 16, up to 40 epochs, and early stopping (patience 8). Augmentation includes random horizontal flip, rotation up to 10° , and random zoom up to 10%.

5.3.3 Results

MobileNetV2 achieves 0.914 validation accuracy, macro F1 0.914, and val log-loss 0.267, with 96 misclassified images (8.6%).

5.4 Modified Scratch CNN

5.4.1 Architecture

The modified CNN retains the three-block architecture and classification head as the scratch CNN, but `BatchNormalization` is replaced throughout with `GroupNormalization(groups=8)`. GroupNorm normalizes within channel groups per sample and carries no running-average statistics, eliminating the train/inference mismatch that caused oscillation in the original scratch CNN at batch size 16.

5.4.2 Training Configuration

All settings match the scratch CNN (Adam 10^{-3} , batch 16, up to 40 epochs, early stopping patience 8, same augmentation) with one addition: `ReduceLROnPlateau` halves the learning rate after 3 epochs without validation loss improvement.

5.4.3 Results

The modified CNN achieves 0.907 validation accuracy and val log-loss 0.249, with 104 misclassified images (9.3%). Training curves are stable throughout, confirming that GroupNorm resolves the oscillation observed in the original scratch CNN. Despite training from scratch with no ImageNet pretraining, the modified CNN is competitive with MobileNetV2 on this dataset.

6 Model Evaluation and Comparison

6.1 Metrics

We report four metrics across all models:

- **Accuracy** — fraction of correctly classified images.
- **Log-loss** — penalizes confident wrong predictions, and captures calibration alongside raw correctness.
- **Macro-averaged F1** — averages per-class F1, which gives equal weight to each class regardless of frequency.
- **Misclassified images** — raw error count on the validation set ($n = 1,120$), reported for interpretability.

6.2 Summary of Results

Table 2 reports validation performance for all models. Full test-set results are reported in Table 3 and discussed in Section 7.

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Table 2: Validation-set performance ($n = 1,120$), sorted by validation loss.

Model	Accuracy	Log-Loss	Macro F1	Errors
Majority Baseline	0.250	—	0.100	840/1120
Logistic Regression	0.846	1.176	0.844	173/1120
Scratch CNN	0.874	0.356	0.877	141/1120
MobileNetV2	0.914	0.267	0.914	96/1120
Modified CNN (GroupNorm)	0.907	0.249	— ^a	104/1120

^a Macro F1 for Modified CNN on validation set is unavailable.

Table 3: Test-set performance ($n = 1,600$).

Model	Accuracy	Log-Loss	Macro F1	Errors
Majority Baseline	0.250	—	0.100	1200/1600
Logistic Regression	0.778	1.958	0.776	356/1600
Scratch CNN	0.811	0.819	0.804	302/1600
MobileNetV2	0.867	0.489	0.863	213/1600
Modified CNN (GroupNorm)	0.852	0.637	0.847	237/1600

6.3 Discussion

Across all models, deep learning methods consistently reduce validation error rates compared to the linear baseline: Logistic Regression achieves an 84.55% accuracy (clears the 25% majority baseline by 59.55 percentage points), the Scratch CNN improves accuracy by a further 2.86 percentage points (reaching 87.4%), and the Modified CNN adds another 3.3 percentage points to reach 90.71%.

The Modified CNN is highly competitive with MobileNetV2 Transfer, which achieves the highest overall validation accuracy at 91.43% and the lowest error rate at 0.0857 (misclassifying only 96 samples compared to the Modified CNN’s 104). This strong performance demonstrates that the original Scratch CNN’s main bottleneck was likely its optimization or normalization choices rather than architecture depth or data volume alone. However, the MobileNetV2 transfer model maintains a slight edge here, outperforming the Modified CNN by 0.72 percentage points in accuracy.

The consistent gap between logistic regression and the CNN models reflects the limits of linear decision boundaries in raw pixel space: while 84.55% validation accuracy is a strong linear result, the glioma–meningioma confusion that dominates errors across all models requires the spatial feature extraction that convolutional layers provide. The validation-to-test drop for logistic regression ($0.8455 \rightarrow 0.777$) is the largest observed, suggesting the linear model overfits pixel co-occurrence patterns specific to the training distribution.

7 Validation and Trustworthiness

7.1 Held-Out Test Set

The 1,600-image test folder was withheld throughout all model development, hyperparameter selection, and architecture decisions. It was evaluated once per model after training was complete. No results from the test set informed any modeling choice.

7.2 Leakage Prevention

The `StandardScaler` is fit on the 4,480-image train-fit split only and applied without refitting to the validation and test sets. Fitting on the full dataset would constitute data leakage and would inflate reported performance. CNN augmentation layers are active only during training and disabled at inference.

7.3 Reporting of Gaps

Validation-to-test accuracy drops are 6.8, 6.3, 4.7, and 5.5 percentage points for logistic regression, scratch CNN, MobileNetV2, and modified CNN respectively. The logistic regression drop is the largest, suggesting some form of linear overfitting. MobileNetV2 shows the smallest drop and is the most generalizable model in this study. The modified CNN, despite leading on validation accuracy and log-loss, drops more than MobileNetV2 on the test set, which suggests that GroupNorm stabilizes training but does not fully prevent overfitting at this scale.

7.4 Reproducibility

All random states are fixed at seed 42 across NumPy, scikit-learn, and TensorFlow. The full training pipeline is contained in a single Colab notebook as well as in a Github Repository; running all cells

in sequence from scratch reproduces all reported results. The team has specifically verified this: when adding new performance metrics without altering the training pipeline, the metrics that were already calculated did not change and this is a physical sign of reproducibility of the results this project produces.

7.5 Prototype Caveat

These results are reported for educational purposes. No clinical conclusions should be drawn. Deployment of any automated diagnostic tool requires prospective clinical validation, patient-level data provenance, demographic diversity testing, and regulatory approval, which are all outside the scope of this project.

8 Error Analysis, Limitations, and Failure Modes

8.1 Glioma–Meningioma Confusion

The dominant misclassification across all models is glioma predicted as meningioma. On the test set, the scratch CNN misclassifies 134 of 400 glioma images as meningioma (33.5%); the modified CNN misclassifies 87 (21.8%); and MobileNetV2 misclassifies 51 (12.8%). The reverse direction, meningioma predicted as glioma, is consistently smaller with 1, 8, and 14 errors respectively. This asymmetric confusion pattern reflects the underlying biology: gliomas arise within brain tissue itself and can appear anywhere in the cerebral hemispheres, and meningiomas grow from the surrounding membranes and press inward. Both, however, can present as large, irregular masses that are difficult to distinguish on a single, high contrast (T1) scan without additional context. Figure 5 shows representative validation errors from the modified CNN.

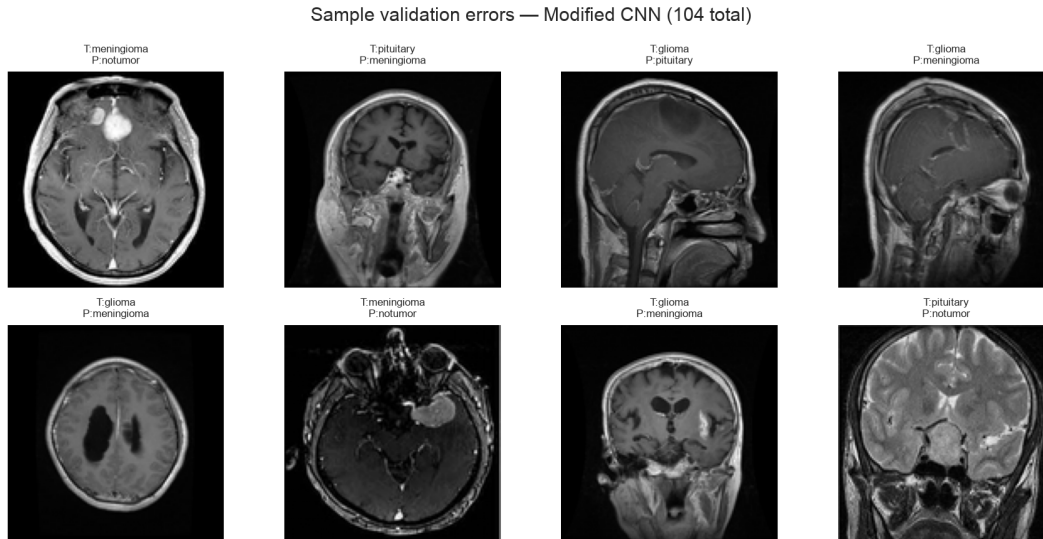
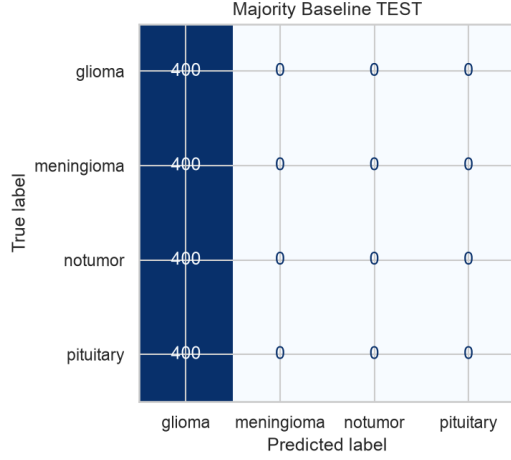
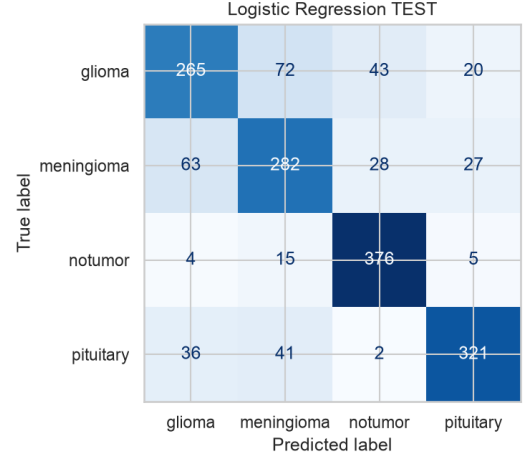


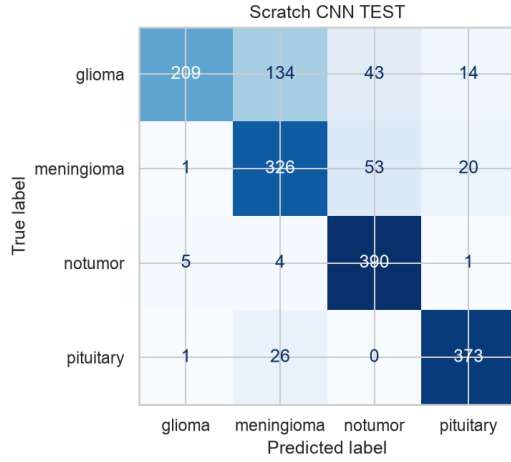
Figure 5: Sample validation errors from the modified CNN (104 total). Each panel shows the true label (T) and predicted label (P). Glioma–meningioma confusion is the dominant error type.



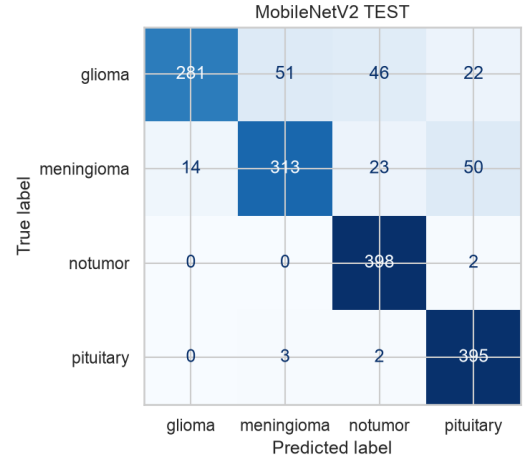
(a) Majority Baseline



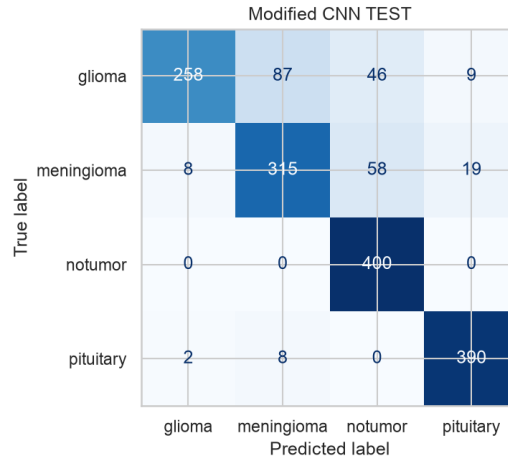
(b) Logistic Regression



(c) Scratch CNN



(d) MobileNetV2



(e) Modified CNN (GroupNorm)

Figure 6: Test-set confusion matrices for all models ($n = 1,600$). The glioma–meningioma off-diagonal entries are the dominant error pattern across all learned models.

8.2 Model-Specific Failure Modes

8.2.1 Logistic Regression

Logistic regression struggles most with glioma, whose variable size, location, and irregular margins produce no consistent pixel-level pattern that a linear boundary can separate from meningioma. The model achieves its strongest performance on notumor scans, where the absence of a mass creates a distinct pixel intensity pattern.

8.2.2 Scratch CNN

The scratch CNN reduces overall error but shows the largest glioma–meningioma confusion of the three deep models, misclassifying 134 glioma as meningioma on the test set. This reflects the BatchNorm instability during training: early stopping may have terminated at a suboptimal checkpoint, leaving the glioma–meningioma boundary underfit.

8.2.3 MobileNetV2

MobileNetV2 achieves the best overall test accuracy and the lowest glioma–meningioma confusion (51 errors). Its primary remaining failure mode is glioma misclassified as notumor (46 errors), suggesting that some glioma presentations are not clearly distinguishable from healthy tissue at the preprocessed 128×128 resolution.

8.2.4 Modified CNN

The modified CNN eliminates notumor errors entirely on the test set (0 notumor misclassified) and achieves perfect classification of the notumor class. Its residual errors are concentrated in glioma (142 total off-diagonal), with 87 glioma predicted as meningioma and 46 as notumor. Meningioma–notumor confusion (58 errors) is also elevated relative to MobileNetV2, suggesting the model trades notumor precision for notumor recall.

8.3 Dataset Limitations

The dataset is sourced from Kaggle and provides no metadata on scanner type, acquisition protocol, or patient demographics. This makes it impossible to assess how well these models would generalize to data collected under different clinical conditions. Without demographic metadata, no subgroup performance analysis is possible. Additionally, only a single MRI modality is used. Additional sequences could improve discrimination between glioma and meningioma specifically, as different sequences highlight different tissue properties. Finally, the dataset provides no patient identifiers, so patient-level independence between the training and test sets cannot be verified, meaning reported metrics may be overly optimistic.

8.4 Scope Limitations

Results are based on a single train–test split with a fixed random seed, so reported metrics carry variance that cross-validation would reduce. Cross-validation was not performed due to computational constraints. These models are educational prototypes developed for AME 34351 and have not been validated for clinical use.

9 Artifact: Streamlit Classification Dashboard

9.1 Design Goals

The artifact allows a user to upload a single MRI image and receive classification probability outputs from all four models simultaneously. All models are evaluated on the same input in a single click and average their confidences to generate a consensus pick. Known limitations are surfaced directly in the interface. Displaying all four models on the same input image allows direct comparison of classical and deep learning outputs on identical data. The inclusion of validation metrics and the glioma–meningioma confusion pattern in the interface itself reinforces that honest reporting of model limitations is as important as maximizing accuracy.

9.2 Interface Description

The dashboard is organized into three panels, shown in Figure 7. The input panel accepts a custom JPEG or PNG upload or allows selection from a set of sample validation images. After uploading, the comparative inference panel displays side-by-side classification probabilities from all four models for the uploaded image. In addition to individual model outputs, the dashboard displays a consensus prediction by averaging the predicted class probabilities across all four models. A third panel displays error analysis and system metadata, including the known glioma–meningioma confusion pattern, dataset provenance, and a validation metrics table comparing all models.

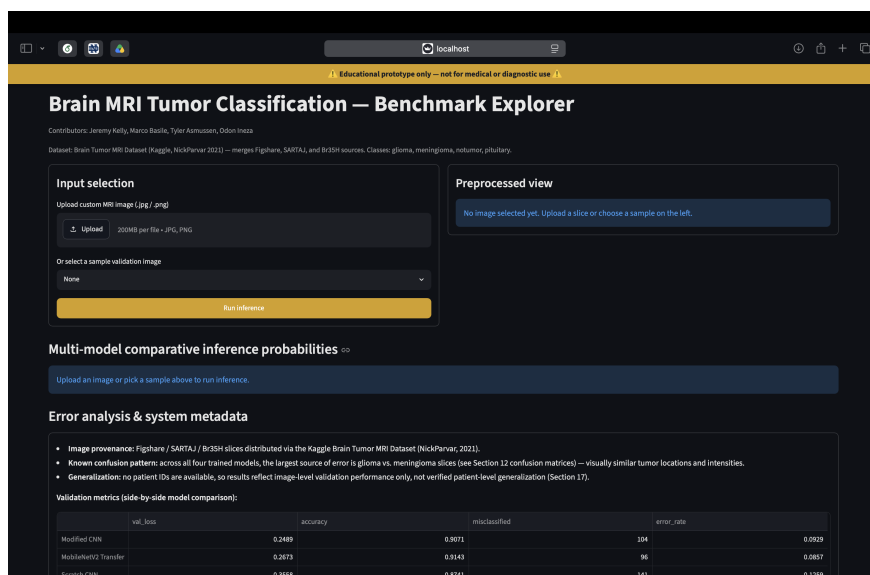


Figure 7: Streamlit classification dashboard. The input panel (left) accepts a custom MRI upload or sample image selection. The lower panel displays validation metrics and system metadata.

9.3 Implementation

The dashboard is implemented in Python using Streamlit. Trained model weights are loaded at startup. Inference runs on the uploaded image after applying the same preprocessing pipeline used during training (grayscale conversion, 128×128 resize, normalization to $[0, 1]$). The logistic regression model is served via scikit-learn; the three CNN models are served via Keras. Because

logistic regression achieves high accuracy but high log-loss (1.176 on validation), its displayed probability outputs may be overconfident and should be interpreted with caution relative to the CNN models. All four models run inference on every uploaded image.

10 Conclusions

10.1 Summary of Findings

This project demonstrates a full ML pipeline applied to a four-class brain tumor MRI classification task, progressing from a 25% majority baseline through logistic regression, a scratch CNN, MobileNetV2 transfer learning, and a modified CNN with GroupNorm. On the held-out test set, MobileNetV2 achieves the strongest generalization at 0.867 accuracy and macro F1 0.863, while the modified CNN achieves 0.852 accuracy despite training from scratch with no ImageNet pretraining. The dominant failure mode across all models is glioma misclassified as meningioma, which is driven by the visual similarity of the two tumor types in single-modality MRI rather than by any correctable modeling choice.

The modified CNN result carries a secondary lesson: the original scratch CNN’s underperformance relative to MobileNetV2 was not due to insufficient architecture depth or data volume, but to BatchNorm instability at small batch sizes. Replacing BatchNorm with GroupNorm closed most of the gap, demonstrating that normalization choice can matter as much as model selection.

10.2 What a Practitioner Should Conclude

These results should be interpreted as a proof of concept, not a clinical benchmark. The models demonstrate that automated MRI classification is tractable on this dataset, but several gaps separate this prototype from anything deployable. Patient-level independence cannot be verified, scanner and acquisition metadata are unknown, only a single MRI modality is used, and the dataset originates from a curated public benchmark rather than a prospective clinical collection. A practitioner should conclude that deep learning approaches are promising for this task, that glioma–meningioma discrimination remains the hardest sub-problem, and that meaningful clinical validation would require prospective data, patient-level splits, multi-modal inputs, and regulatory review.

10.3 Future Work

Several directions could meaningfully improve on these results. Multi-modal input, which involves combining T1 with T2 or FLAIR sequences, would provide tissue contrast information that single-modality classification cannot access, and would likely reduce glioma–meningioma confusion specifically. Tumor segmentation prior to classification would focus the model on lesion regions rather than whole-image crops, removing background anatomy as a confounding factor. Cross-validation across multiple train–test splits would produce more reliable performance estimates than the single split reported here. Finally, fine-tuning on a clinically sourced dataset with verified patient-level splits and demographic metadata would be the necessary next step before any deployment could be considered.

10 References

- [1] Venkatesan Rajinikanth, Seifedine Kadry, and Yunyoung Nam. Convolutional-neural-network assisted segmentation and SVM classification of brain tumor in clinical MRI slices. *Information Technology and Control*, 50(2):342–356, 2021. ISSN 2335-884X. doi: 10.5755/j01.itc.50.2.28087.
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- [3] Masoud NickParvar. Brain tumor MRI dataset. Kaggle, 2021. URL <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>. Accessed: June 2026.
- [4] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520, 2018. doi: 10.1109/CVPR.2018.00474.

A Hyperparameter Summary

Table 4: Hyperparameter summary across all models.

Model	Parameter	Value
Logistic Regression	C	1.0
	Solver	saga
	Max iterations	1,000
Scratch CNN	Filters	32, 64, 128
	Dense head	64 units, ReLU
	Dropout	0.3
	Normalization	BatchNorm
	Optimizer	Adam (10^{-3})
	Epochs / batch	40 / 16
MobileNetV2	Backbone	ImageNet weights
	Frozen layers	All (Stage 1)
	Fine-tuned layers	Last 50 (Stage 2)
	Stage 1 LR	10^{-4}
	Stage 2 LR	10^{-5}
	Dropout	0.2
	Epochs / batch	40 / 16
Modified CNN	Filters	32, 64, 128
	Dense head	64 units, ReLU
	Dropout	0.3
	Normalization	GroupNorm (groups=8)
	Optimizer	Adam (10^{-3})
	Epochs / batch	40 / 16
All models	Image size	128×128
	Pixel range	[0, 1]
	Random seed	42

B Repository and Deployment

The full training pipeline is available at <https://github.com/keremyjelly/tumor-classification>. The Streamlit dashboard can be run locally with:

```
pip install -r requirements.txt
streamlit run app.py
```

The Colab notebook containing all training code and evaluation cells is provided in the repository.